

EXECUTIVE SUMMARY

CO₂ Storage Assessment

Executive Summary — Preliminary Screening

FORMATION

Alberta Basin

LOCATION

Alberta, Canada

PREPARED FOR

Not specified

Applicant / Client

PREPARED BY

TODAK2000

CarbonLens Simulation Studio v3

DATE ISSUED

12 June 2026

Ref: CL-EXEC-20260612

PRELIMINARY SCREENING

Not for regulatory submission · Independent review required



✓ DIGITAL TWIN REGISTRY — PENDING REVIEW

Alberta Basin CO₂ Geological Storage Certificate

Asset ID: CL-002F04-14-1000 Stored: 135.79 Mt CO₂ Containment: 78% Safety Factor: 1.302 Jurisdiction: CA

Share: <https://carbonlens.app/registry/verify/CL-002F04-14-1000> — Issued 12 June 2026 by CarbonLens Studio v3

CO₂ STORED

135.79

Mt — Year 50 of 50

P50 CAPACITY

93.2

Mt — P90 23.8 / P10 256.4 Mt

SAFETY RATING

MODERATE

SF: 1.30

CONTAINMENT

78%

Plume containment probability

Storage Capacity Analysis

Parameter	Value
Stored CO ₂	135.79 Mt
P50 Capacity	93.22 Mt
P10 (optimistic)	256.36 Mt
P90 (conservative)	23.77 Mt
Capacity Utilisation	145.5%
Storage Efficiency (Cc)	2.00%

DOE CAPACITY RANGE (Mt)

P90 (cons.) 23.8 P50 93.2 P10 (opt.) 256.4



■ Injected vs P50 capacity

TRAPPING MECHANISM DISTRIBUTION

Residual		39.890 Mt (29.4%)
Dissolved		40.498 Mt (29.8%)
Mineral		0.302 Mt (0.2%)
Mobile		55.098 Mt (40.6%)

Fluid Properties

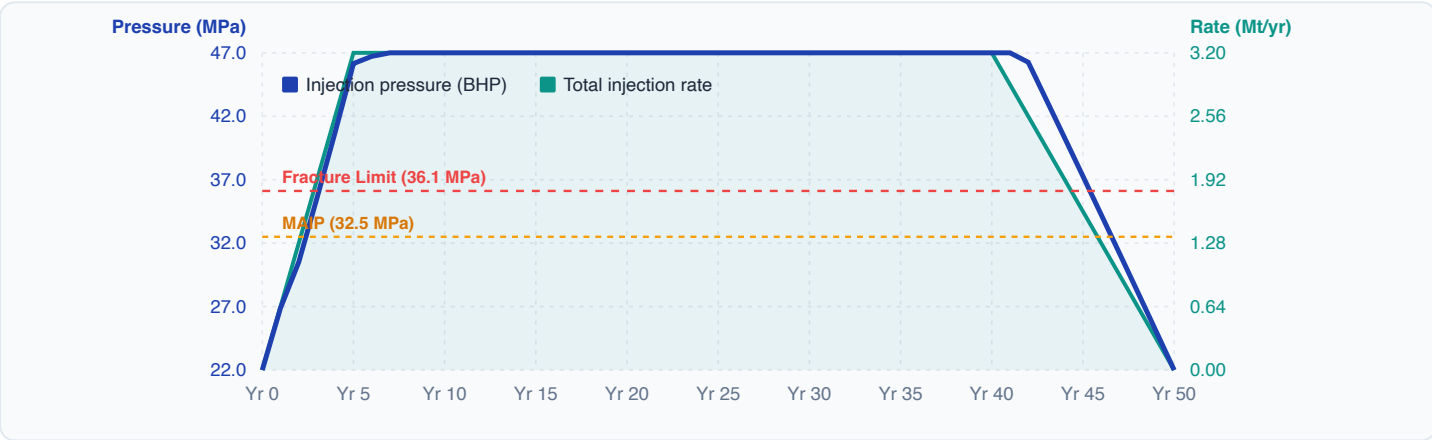
Property	Value
CO ₂ Density	712.7 kg/m ³
CO ₂ Viscosity	0.0534 mPa·s
Brine Density	1088.6 kg/m ³
CO ₂ Solubility	0.4734 mol/kg
Diffusion Coefficient	1.845e-10 m ² /s
Interfacial Tension (IFT)	33.38 mN/m
Phase State	SUPERCRITICAL
Injection Pressure	25.08 MPa

Geomechanical Assessment

Parameter	Value	Assessment
Safety Factor	1.302	MARGINAL

Parameter	Value	Assessment
Fracture Pressure (Hubbert-Willis)	36.12 MPa	REF
Min. Horizontal Stress (σ_h)	36.12 MPa	REF
Caprock Stress	24.41 MPa	REF
MAIP	32.50 MPa	+7.42 MPa headroom
Surface Heave	0.0 mm	ACCEPTABLE
Seismicity Risk	MODERATE	MODERATE
Mohr-Coulomb Margin	13.662	PASS

INJECTION PRESSURE & FLOW RATE PROFILE



Fracture Mechanics & Storage Integrity Notice: If injection pressure exceeds the rock fracture limit, tensile fracturing degrades containment by creating vertical leakage pathways and pressure bypass zones. This directly compromises fluid flow assurance (due to bypass paths and loss of sweep efficiency) and restricts active storage capacity by capping the safe operating bottomhole pressure (BHP). Keeping pressures below the Maximum Allowable Injection Pressure (MAIP) is mandatory for storage permit validation.

 MONTE CARLO PROBABILISTIC BOUNDS

500 LHS realizations • Alberta Basin

P10 (Conservative)
69.23
Mt CO₂

P50 (Expected)
89.50
Mt CO₂

P90 (Optimistic)
115.97
Mt CO₂

P90/P10 Spread
1.7×
ratio

Peak ΔP P50
22.00
MPa

Uncertainty assessment: Low — well-constrained

Sampled: permeability ±30%, porosity ±3.0%, area ±20%, thickness ±15%. DOE Goodman (2011) volumetric method.

Injection Well Programme

Well ID	Rate (Mt/yr)	Ramp-up (yr)	Ramp-down (yr)	Project Duration (yr)
Well 1	1.195	5	10	50
Well 2	1.000	5	10	50
Well 3	1.000	5	10	50
TOTAL	3.195 Mt/yr			

Project Economics & Incentives

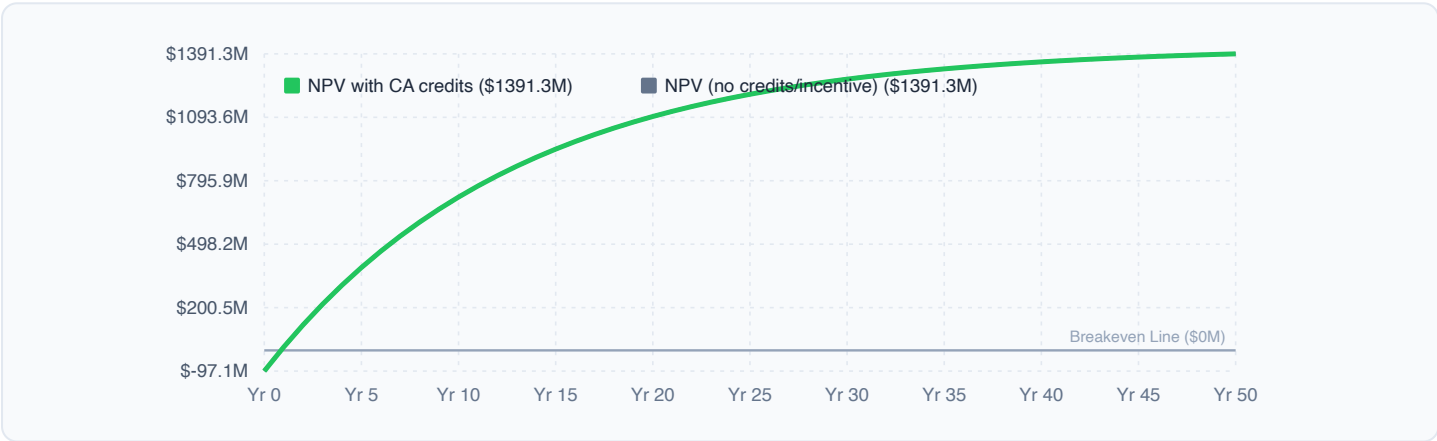
FINANCIAL SUMMARY

Metric	Value
Jurisdiction Carbon Pricing	\$48/t (CA)
Total CAPEX	\$97.1M
OPEX (per tonne)	\$3.20/t
Total OPEX	\$434.5M
Levelized Cost (LCO ₂ S)	\$3.92/t
Net Cost after Credits	\$-44.08/t

CAPEX BREAKDOWN

Drilling		\$54.6M (56%)
Facilities		\$11.0M (11%)
Pipeline		\$26.3M (27%)
Monitoring		\$5.2M (5%)

PROJECT CUMULATIVE NET PRESENT VALUE (NPV) PROFILE

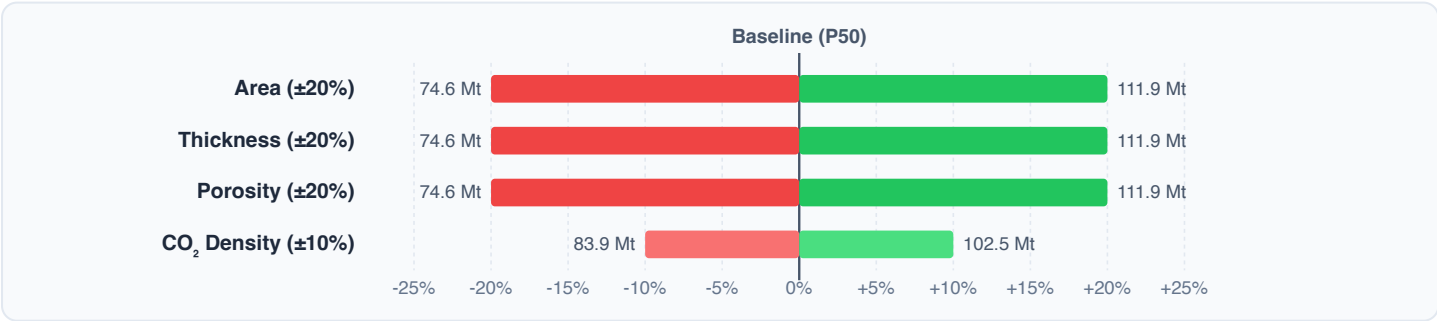


Project Duration is the full injection period (from UIStore). Cumulative injection accounts for ramp-up and ramp-down schedules within this period. Simple rate × duration gives a conservative upper bound; actual cumulative injection is lower when ramp periods are present.

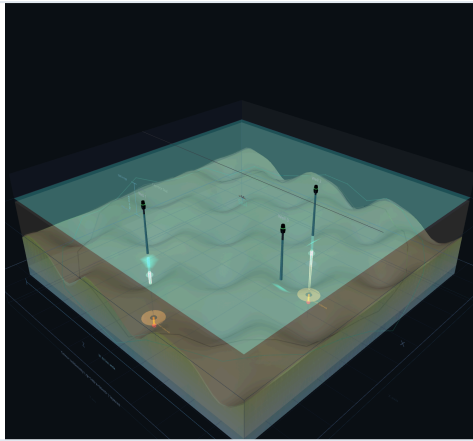
Storage Capacity Sensitivity

Shows the dynamic impact of a ±20% shift in key reservoir parameters (Thickness, Porosity, Area) and a ±10% shift in CO₂ Density on the P50 storage capacity estimate.

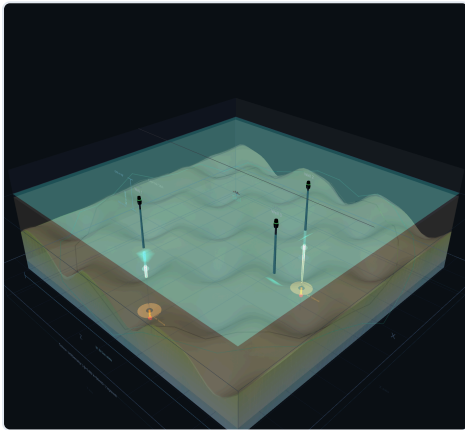
P50 CAPACITY SENSITIVITY TORNADO CHART (BASE: 93.2 MT)



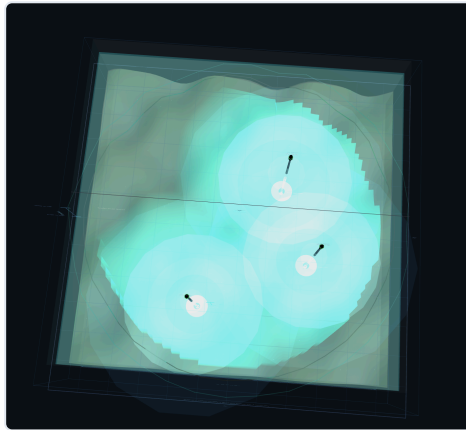
Reservoir Evolution



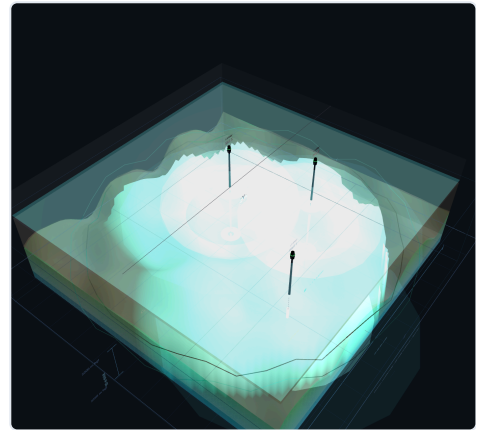
Year 1 — Plume State



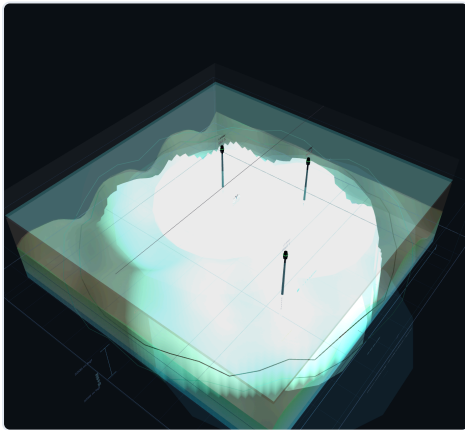
Year 1



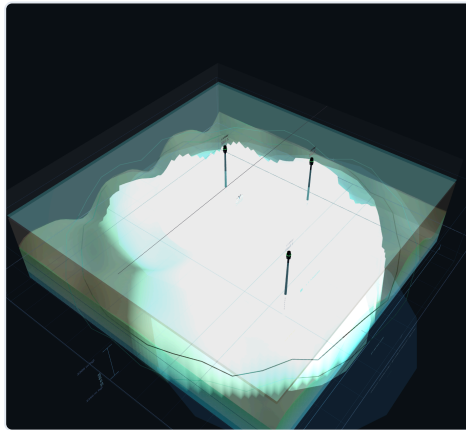
Year 10



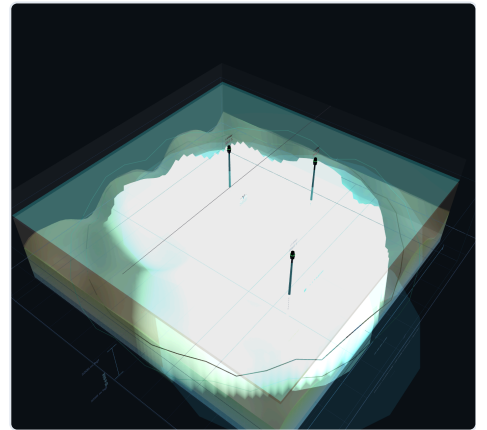
Year 20



Year 30



Year 40



Year 50

Recommendations

1. Advance to detailed site-specific characterisation and full-physics reservoir simulation to confirm screening estimates before regulatory submission.
2. Capacity utilisation of 145.5% exceeds 80% threshold. Reduce injection rate or increase area of review to manage overpressure risk.
3. Safety factor of 1.30 is below the recommended 1.5 minimum. Reduce injection pressure or improve caprock characterisation.
4. Containment probability of 78% supports advancing to front-end engineering study and regulatory pre-application screening.
5. **For Engineers:** Commission core-flooding laboratory tests on reservoir samples to construct field-validated relative permeability (k_{rg} , k_{rw}) and capillary pressure curves under in-situ conditions.
6. **For Regulators:** Define a localized Monitoring, Reporting, and Verification (MRV) program (incorporating active distributed temperature sensing (DTS), microseismic arrays, and deep groundwater sampling) to satisfy Class VI / EU Directive compliance.
7. **For Investors:** Mitigate long-term overpressure risk by optimizing the injection well pattern (horizontal well completions or multi-well distribution) to increase active sweep efficiency and maximize capacity utilization.
8. All capacity estimates are preliminary screening values using DOE Goodman et al. (2011) methodology. Regulatory-grade estimates require site-specific characterisation.

Equation & Model Reference

Module / Engine	Equation / Correlation	Reference
CO ₂ EOS	$\rho(P,T) = PM/ZRT$; Z from Peng–Robinson cubic EOS; $\kappa(\omega) = 0.37464 + 1.54226\omega - 0.26992\omega^2$	Peng & Robinson (1976), Ind. Eng. Chem. Fundam. 15(1):59–64; $T_c=304.13$ K, $P_c=7.377$ MPa, $\omega=0.2239$. <i>Note: Span-Wagner (1996) 56-term Helmholtz EOS is the upgrade target for $\pm 0.5\%$ accuracy.</i>
CO ₂ Viscosity	$\mu(T,\rho) = \mu_0(T) + \Delta\mu(T,\rho)$; zero-density + excess; $\epsilon/k=251.196$ K	Fenghour, Wakeham & Vesovic (1998), J. Phys. Chem. Ref. Data 27(1):31–44. <i>Upgrade target: Laesecke & Muzny (2017), JPCRD 46:013107 (NIST reference, $\pm 0.5\%$ vs $\pm 5\%$ in supercritical region).</i>
CO ₂ Solubility	$\ln(y_2 \cdot P) = \mu^0(T) + 2\lambda(T,P) \cdot m + \xi(T,P) \cdot m^2$; Pitzer ionic strength	Duan & Sun (2003), Chem. Geology 193:257–271
CO ₂ -Brine IFT	$\sigma = \text{MARS}(P, T, S)$; multivariate adaptive regression splines trained on experimental data	Olagunju (in prep.), MSc UTP Malaysia; Chiquet et al. (2007), Energy Convers. Mgmt. 48:736
Storage Capacity	$M = A \cdot h \cdot \phi \cdot \rho_{CO_2} \cdot S_g \cdot E_{eff}$; volumetric DOE methodology	Goodman et al. (2011), Int. J. GHG Control 5(4):853–866; DOE (2010) Atlas
Residual Trapping	$S_{gr} = S_{gi} / (1 + C \cdot S_{gi})$ (Land, 1968); $V_r = \phi \cdot (1-S_{wi}) \cdot S_{gr} \cdot A \cdot h \cdot \rho_{CO_2}$	Pentland et al. (2011), SPE-133798; Land (1968), Trans. AIME 243:149
Solubility Trapping	$m_{diss} = \phi \cdot S_w \cdot \rho_{brine} \cdot X_{CO_2}(T,P,S) \cdot A \cdot h$; aqueous dissolution	Duan & Sun (2003); Ennis-King & Paterson (2005), SPE-88502
VE Plume Migration	$\partial(h_g/B)/\partial t + \nabla \cdot Q_g = Q_{inj}/B$; vertical–equilibrium thin-lens approximation	Nordbotten & Celia (2006), Water Resour. Res. 42:W01207
Pressure / AoR	$\Delta P(r,t) = Q/(4\pi kh)[\mu_b E_1(r^2/4\alpha_b t) + (\mu_{eff}-\mu_b)E_1(R_p^2/4\alpha_b t)]$; two-phase composite; $\mu_b=\text{brine}$, $\mu_{eff}=\mu_{CO_2}/k_r$	Nordbotten, Celia & Bachu (2005), Transp. Porous Media 58(3):339–360; Theis (1935) as single-phase baseline; Hantush & Jacob (1955) for leaky-aquifer option
Rel. Permeability	$k_{rg} = k_{rg}^0 \cdot (1-S_e)^{\frac{1}{2}} \cdot (1-S_e^{1/m})^{2m}$; $k_{rw}=k_{rw}^0 \cdot S_e^{\frac{1}{2}} \cdot (1-(1-S_e^{1/m})^m)^2$; $m=1-1/n$; + Killough–Land hysteresis	van Genuchten (1980), Soil Sci. Soc. Am. J. 44:892; Mualem (1976), WRR 12:513; Krevor et al. (2012), WRR 48:W02514; Land (1968), Trans. AIME 243:149; Killough (1976), SPE J. 16(1):37
Fracture Pressure	$P_f = [\nu/(1-\nu)](\sigma_v-P_h) + P_h + T_0$; minimum horizontal stress criterion	Hubbert & Willis (1957), Trans. AIME 210:153–166
Mohr-Coulomb	$\tau = c + \sigma_n \tan(\phi)$; $SF = \tau_{strength}/\tau_{drive}$; failure when $SF < 1$	Jaeger, Cook & Zimmerman (2007), Fundamentals of Rock Mechanics, 4th ed.
MAIP	$MAIP = 0.9 \cdot P_f - P_{res}$; maximum allowable injection pressure	Bachu et al. (2007), Int. J. GHG Control 1(4):374; EPA UIC Class VI §146.88
Surface Heave	$\delta \approx (\Delta P \cdot h) / E \cdot (1-\nu^2) \cdot a$; elastic nucleus-of-strain approximation	Teatini et al. (2011), J. Geophys. Res. 116:B08204
Area of Review	$r_{AoR} = \sqrt{(4Kt/\phi\mu c_t)}$; pressure perturbation front ≥ 6.9 kPa threshold	Chabora & Benson (2009), GHGT-9; EPA Class VI UIC Program Guidance

| About CarbonLens

CarbonLens is a browser-based CO₂ geological storage simulation studio, purpose-built for deep saline aquifer CCS screening. It delivers real-time plume simulation, geomechanical risk assessment, and regulatory permit preparation — entirely in the browser, with no server or proprietary software required.

Developed as MSc research at Universiti Teknologi PETRONAS (UTP), Malaysia, in partnership with the CarbonLens product team.

RESEARCH TEAM

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SIMULATION ENGINE

Peng-Robinson EOS · DOE Goodman (2011) · Duan-Sun Solubility · MARS IFT (Olagunju, in prep.) · Nordbotten (2005) Two-Phase AoR · Mohr-Coulomb Geomechanics · van Genuchten–Mualem kr · Killough–Land Hysteresis

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